

Employee Performance and Workforce Analysis

A Comprehensive Data Analysis Report

Prepared from cleaned_messy_employees dataset

June 2026

1. Project Overview

This project analyzes employee performance and workforce dynamics using HR data from 1,020 employees across various departments and regions. The goal is to uncover insights into salary patterns, performance distribution, remote work trends, and workforce segmentation to guide strategic HR decisions.

2. Dataset Summary

- Rows: 1,020
- Columns: 11

Key Features:

- Employee demographics (Employee ID, First Name, Last Name, Age, Age Group)
- Organizational details (Department, Region, Employment Status, Hire Date)
- Performance & compensation (Salary, Performance Rating, Remote Work)

Missing Data: No missing values in the cleaned dataset (null values were handled during preprocessing).

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading** – Imported the dataset using pandas.
- **Initial Exploration** – Used `df.info()` to check structure and `.describe()` for summary statistics.

	count	mean	std	min	max
age	1020	31.97	5.14	25	40
salary	1020	85,173.29	19,659.19	50,047	119,972

- **Missing Data Handling** – Verified no null values remained in the cleaned dataset; original messy data had inconsistencies resolved during preprocessing.
- **Column Standardization** – Renamed columns to snake_case for better readability and documentation.
- **Feature Engineering** –
 - Created age_group column by binning employee ages into Young-Adult, Adult, and Middle-Aged.

- Parsed department_region into separate department and region fields for granular analysis.
- **Data Consistency Check** – Verified remote_work field (t/f) was consistent across all records; standardized boolean representation.
- **Dashboard Integration** – Connected cleaned dataset to Power BI and built an interactive dashboard for visual insights.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis to answer key workforce business questions:

1. **Salary by Department** – Compared total and average salary across all six departments.
2. **High-Earning Remote Workers** – Identified remote employees earning above the average salary of \$85,173.
3. **Top 5 Departments by Average Salary** – Found department-region combinations with the highest average compensation.

department_region	avg_salary	count
Admin-Illinois	92,416.79	29
Cloud Tech-Florida	92,126.14	28
Sales-Florida	90,582.70	33
DevOps-Nevada	90,001.74	27
Admin-California	89,806.06	34

4. **Remote Work Comparison** – Compared average salary between onsite (False: \$85,851.90) and remote (True: \$84,502.62) employees.
5. **Performance vs. Status** – Checked whether Active employees are more likely to have Excellent or Good performance ratings.

status	avg_salary	total_employees	active_top_perf
Active	85,105.31	352	195
Pending	85,698.69	356	—
Inactive	84,650.50	312	—

6. **Discount-Dependent Departments** – Identified departments where over 50% of employees work remotely.
7. **Employee Segmentation** – Classified employees into age groups — Young-Adult, Adult, and Middle-Aged — based on age ranges.

age_group	number_of_employees
Young-Adult	622
Middle-Aged	210
Adult	188

8. **Top Departments per Region** – Listed the most populated department-region combinations.
9. **Repeat Performance & Remote Work** – Checked whether remote employees are more likely to have higher performance ratings.
10. **Salary by Age Group** – Calculated average salary contribution of each age group.

age_group	avg_salary
Young-Adult	84,323.85
Middle-Aged	88,353.27
Adult	84,431.59

5. PostgreSQL Connection

```
from sqlalchemy import create_engine username = "postgres" passowrd = "030822"
host = "localhost" port = "5432" database = "messy_employee" engine = create_en
gine(f"postgresql+psycopg2://{username}:{passowrd}@{host}:{port}/{database}")
```

A database called “messy_employee” was already created in PostgreSQL, which is further connected to the Notebook. The PostgreSQL establishes connection with Notebook.

6. Load Data to PostgreSQL

```
table_name = "employee" df.to_sql(table_name, engine, if_exists="replace",
index=False) print(f"Data has been successfully loaded into the table
'{table_name}' in database '{database}'.")
```

Further, the **Clean Dataset** is loaded into the “employee” table of the “messy_employee” database in PostgreSQL.

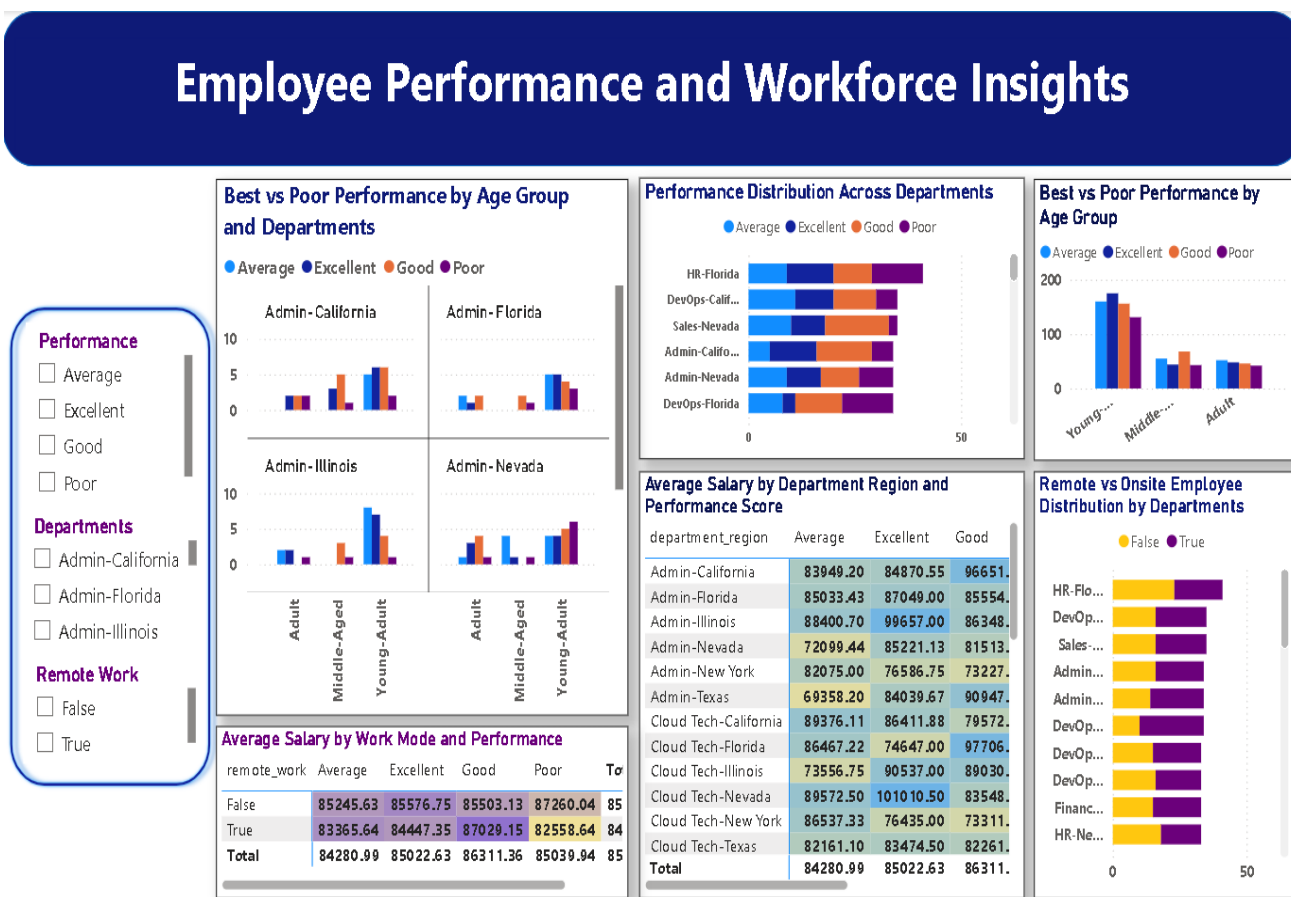
7. Dashboard in Power BI

Finally, we built an interactive dashboard in Power BI to present workforce insights visually.

The dashboard includes the following panels:

- Best vs Poor Performance by Age Group and Departments — clustered bar chart breakdown
- Performance Distribution Across Departments — horizontal stacked bar chart
- Best vs Poor Performance by Age Group — summary bar chart
- Average Salary by Department Region and Performance Score — cross-tab matrix
- Average Salary by Work Mode and Performance — onsite vs. remote comparison table
- Remote vs Onsite Employee Distribution by Departments — horizontal bar chart

Interactive filters on the left panel allow slicing by Performance rating, Department, and Remote Work status.



8. Business Recommendations

- **Performance-Based Compensation** – Review salary bands for employees rated 'Good' and 'Excellent' — currently the gap versus 'Average' is narrow; stronger differentiation can motivate performance.
- **Remote Work Policy** – Onsite employees earn marginally more on average (\$85,851 vs \$84,502). Evaluate whether remote work is equitably compensated and standardize the policy.
- **Age Group Investment** – Middle-Aged employees command the highest average salary (\$88,353) but represent only 20.6% of the workforce; succession planning should account for this gap.
- **Departmental Focus** – Admin-Illinois and Cloud Tech-Florida report the highest average salaries; benchmark these against performance outputs to confirm ROI.
- **Workforce Status Review** – Pending employees (35%) outnumber Active (34%) — a high pending ratio may signal onboarding bottlenecks that should be addressed to improve productivity.